Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory (ADF)

Objective

The objective of this assignment was to understand the basic concepts of Microsoft Azure Cloud and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF).

Azure Services Used

- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)
- Linked Services
- Source and Destination Datasets
- Get Metadata Activity
- Copy Data Activity
- IAM (Identity and Access Management)

Implementation Steps

1. Created a **Resource Group** to organize all Azure resources.
2. Created a **Storage Account** and a Blob Container to store the dataset.
3. Uploaded the **Superstore CSV file** to the Blob Storage container.
4. Created an **Azure Data Factory** instance and explored the Author, Monitor, and Manage sections.
5. Created a **Linked Service** to connect Azure Data Factory with Azure Blob Storage.
6. Created **Source** and **Destination Datasets** for reading and writing the CSV file.

7. Added the **Get Metadata** activity to validate the source file before processing.
8. Added the **Copy Data** activity to copy data from the source container to the destination container.
9. Executed the pipeline using **Debug** and **Trigger Now** options.
10. Monitored the pipeline execution and verified that the data was copied successfully.
11. Assigned the required **Reader** and **Contributor** IAM roles to enable secure access between Azure Data Factory and Azure Storage.

Pipeline Flow

Source CSV File (Blob Storage) → Get Metadata → Copy Data Activity → Destination Blob Storage

Output

The pipeline successfully copied the Superstore dataset from the source Blob Storage container to the destination container. The Get Metadata activity validated the source file before execution, and the pipeline execution status was successfully monitored in Azure Data Factory.

Conclusion

This assignment provided practical knowledge of Microsoft Azure services, including Azure Storage, Blob Containers, Azure Data Factory, Linked Services, Datasets, Get Metadata, Copy Data Activity, IAM Roles, and the implementation of an end-to-end ETL data pipeline.

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